CS2204 Fundamentals of Internet Application Development
Course Work No. 1 (CW1)
Semester A 2025-2026

10% Marks
Due Date: 10 October 2025, 23:59 [Week 6]

Learning Outcomes:

- Design structures of web pages for a more realistic, commercial-like website
- Achieve user requirements by using appropriate HTML5 markups.
- Produce web pages that can pass through validation.

1. Overview

You must build a simple Web site for Intelligent Sensing Lab by going through the 3S's: structure, style, and script. This CW1 is the first step focusing on structure only. The Web site, therefore, would not be fully functional until the second and third parts (CW2 & CW3) using CSS and JavaScript are completed.

To encourage you to have critical thinking and investigation, only the basic information of the Web pages is specified, and you have the freedom to design the structure if it can fulfill the specified requirements. However, like any design, some structures are better than others.

To ensure timely feedback to the student's questions, each of the following TAs will be responsible for different groups of students according to **the session of the tutorial**. If you have any questions, you can contact the corresponding TA directly by email. Each TA will also attend the corresponding tutorials in person to answer your questions related to the assignment.

- **T01/TA1(Tue. 15:00-16:00): MAHAD Muhammad** (mmahad2-c@my.cityu.edu.hk)
- **T02(Mon. 14:00-15:00): DING Xiaoyu** (xiaoyding6-c@my.cityu.edu.hk)
- **T03(Mon. 12:00-13:00): YUE Zhaoxi** (zhaoxiyue2-c@my.cityu.edu.hk)
- **T04(Wen. 10:00-11:00): FU Qianli** (qianlifu2-c@my.cityu.edu.hk)
- **T05(Wen. 17:00-18:00): FENG Yuxiang** (yuxiafeng2-c@my.cityu.edu.hk)

Guideline for late submissions:

- Late within **3 hours - 10% marks deduction**
- Late within **3 ~ 12 hours - 50% marks deduction**
- After **12 hours - no** submission will be accepted

2. Requirements of Structure

Required information on the Web site is grouped into blocks, then pages. There are six main pages and one design page. All contents, unless specified, should be completed by yourself. If you use information and ideas from others online, you **MUST** acknowledge and quote the source on the design page. **If you are found to have plagiarized, you will be subject to severe penalties, including but not limited to a zero for this coursework, failing the course, and being reported to the school authorities. Therefore, please follow the above rules to ensure the successful pass this course.**

There are 7 pages in total. The default landing page of the Web site is the *Home* page. The contents of every page can be found in the file "cw1-contents.txt" available on Canvas.

Home Page

a) This page contains information grouped into 5 blocks, 3 of which are the standard header, navigation bar, and footer blocks as follows:

a.1) The header/banner block consists of a logo (**logo.png**) and the heading of the Website; these are separated and should not be done in one single image. You should use **<header>** for the heading block. The size of the logo should be set to **550×150**. The following is an example of how to set the size of an image:

<img src="…" width="550" height="150">

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a.2) The navigation bar should include links to all the website's six main pages (except the Design page); all jump hyperlinks must be valid in the navigation bar. Ensure all links point to the correct page so users can access the required information without problems. Also, ensure that the layout and design of the navigation bar are consistent across all pages so that users can easily find the information they need. You also need to note that you need to use a reasonable HTML structure to organize the navigation bar. You need to note that there is no requirement for the navigation bar to be horizontal or vertical. **<nav>** should be used for the navigation bar.

- Home
- About
- Apply
- Visit
- Information
- Contact

a.3) The footer with another logo (**footer_logo.png, 460×100**) and information about the copyright and a link pointing to the Design page. **<footer>** should be used for the footer block.

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Please choose a proper way to display the copyright icon and replace “CS2204” by your name with a link to the *Design* page.

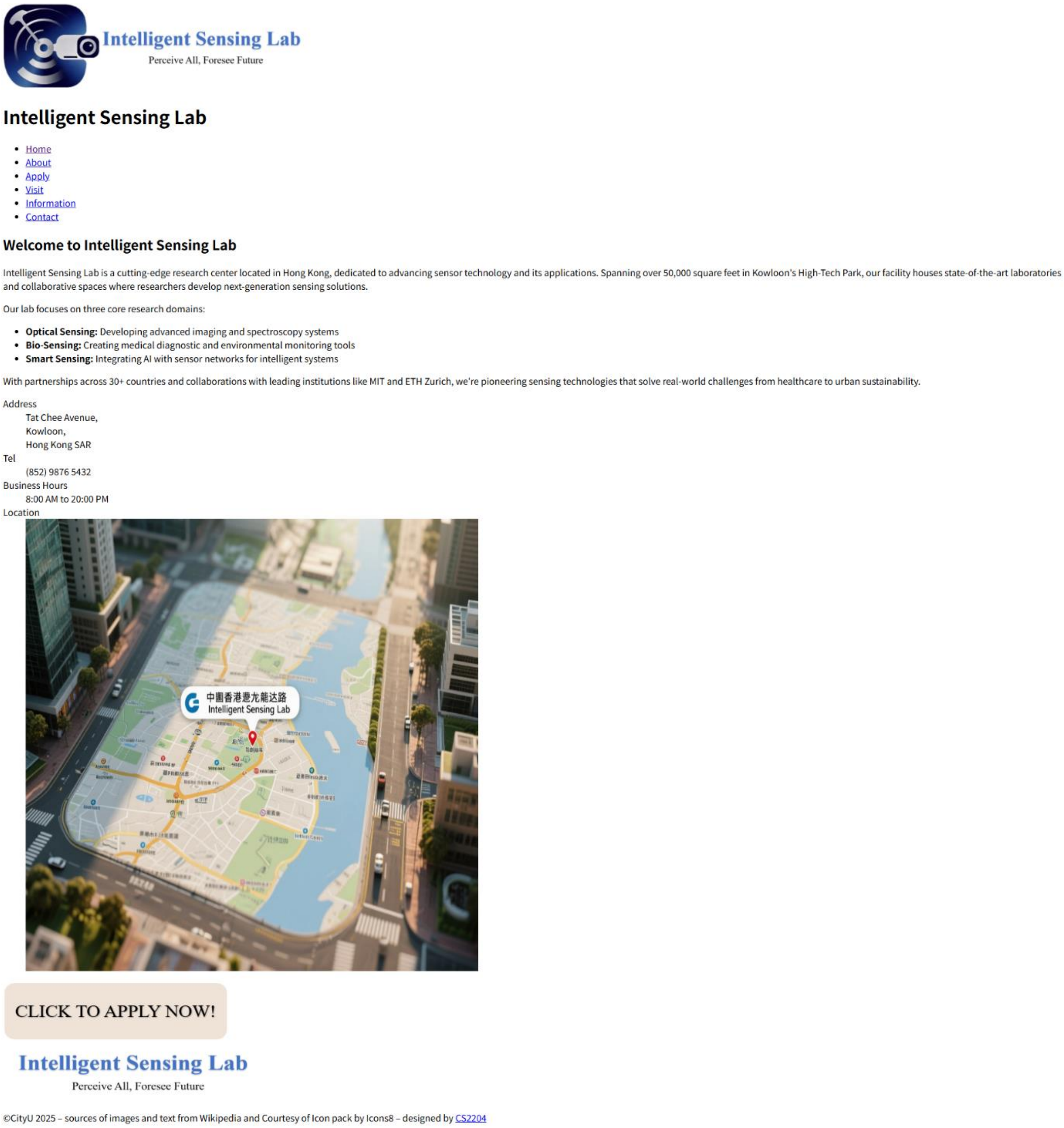
These three blocks would appear on not only the home page but also all other pages. HTML code of them should be duplicated on every page.

b) The remaining 2 blocks are as follows:

b.1) The welcome block contains some information shown in the screenshot. You can find the content of this page in the appendix at the end of the document. The website we needed to implement had a large amount of text, so we provided all the website’s textual content in the appendix.

b.2) The basic information block contains four types of information shown in the screenshot: address, telephone, business hours, and location which is an image (**location.png, 800×800**); use the most structured elements to present such information. At the end, an image (**button.png, 400×100**) link points to the Apply page. We have implemented the button’s functionality with an image here.

The following is an example of a finished home page.

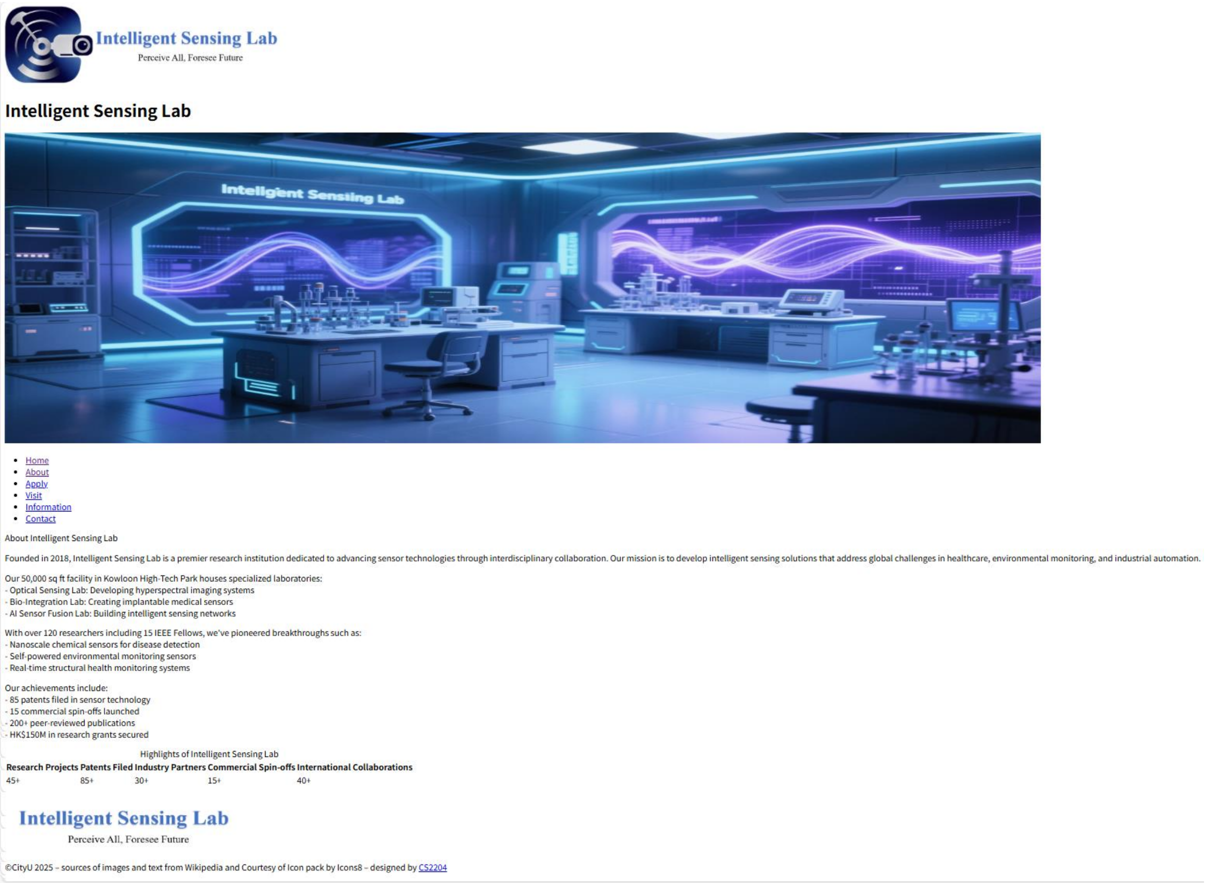


About Page

c) contains the standard header, navigation bar, and footer, the same as the *home* page. The About page also includes two more blocks: the **promotional image block** and the **information block**.

d) You should put an extra **promotional image block** inside the header that contains only one image (**about.png, width="2000", height="600"**). You should put it in the <header> with the header block introduced before.

e) **The information block** should display the information below. Use <p> for each paragraph. Also, it should display the "Highlights of Intelligent Sensing Lab" table that contains some brief information. The cw1 does NOT require the "Highlights of Intelligent Sensing Lab" table's border.



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About Intelligent Sensing Lab

Founded in 2018, Intelligent Sensing Lab is a premier research institution dedicated to advancing sensor technologies through interdisciplinary collaboration. Our mission is to develop intelligent sensing solutions that address global challenges in healthcare, environmental monitoring, and industrial automation.

Our 50,000 sq ft facility in Kowloon High-Tech Park houses specialized laboratories:
- Optical Sensing Lab: Developing hyperspectral imaging systems
- Bio-Integration Lab: Creating implantable medical sensors
- AI Sensor Fusion Lab: Building intelligent sensing networks

With over 120 researchers including 15 IEEE Fellows, we've pioneered breakthroughs such as:
- Nanoscale chemical sensors for disease detection
- Self-powered environmental monitoring sensors
- Real-time structural health monitoring systems

Our achievements include:
- 85 patents filed in sensor technology
- 15 commercial spin-offs launched
- 200+ peer-reviewed publications
- HK$150M in research grants secured

Highlights of Intelligent Sensing Lab

Research Projects	Patents Filed	Industry Partners	Commercial Spin-offs	International Collaborations
45+	85+	30+	15+	40+

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Apply Page

f) It contains the standard header, navigation bar, and footer, the same as the *home page*. The order page also includes three more blocks.

g) **The welcome block** contains welcome information. On this webpage, we include the phrase "Welcome to ISLab" in the block.

The apply block shows the groups in **Intelligent Sensing Lab** that can be applied; it has the heading “Join our cutting-edge research teams in these focus areas:” followed by three more minor headings with names (Optical Sensing Division, Bio-Sensing Division, and Smart Sensing Division). You should design three research group application forms for the Optical Sensing Division (Hyperspectral Imaging Group, LiDAR Development Team, Nanophotonics Research), three research group application forms for the Bio-Sensing Division (Medical Diagnostics Group, Environmental Sensors Team, Wearable Technology Lab), and two group application forms for the Smart Sensing Division (IoT Sensor Networks, AI-Enabled Sensing)**.**

Each research group is a <form> including a <img> **(200×200)**, a <label>, an <input>, and a <button>. We do not use these buttons in this coursework. We will functionalize them in later coursework with JS. Companies in the same division should be organized in a <div>. The “action” and “method” of each form should be set to “#” and “get”, respectively. **The resulting block** displays information about chosen companies, rankings, and the number of completion choices in a table format. The head of this table, the deadline, and “Total number of completed choices …” are three sub-headings like <h3>.

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Welcome to ISLab

Join our cutting-edge research teams in these focus areas:

Optical Sensing Division

Hyperspectral Imaging Group [ ] rank of choice

LiDAR Development Team [ ] rank of choice

Nanophotonics Research [ ] rank of choice

Bio-Sensing Division

Medical Diagnostics Group [ ] rank of choice

Environmental Sensors Team [ ] rank of choice

Wearable Technology Lab [ ] rank of choice

Smart Sensing Division

IoT Sensor Networks [ ] rank of choice

AI-Enabled Sensing [ ] rank of choice

Your chosen research groups:

Current Internship Placements:

Division	Research Group	Rank
Optical Sensing	Hyperspectral Imaging Group	1
Bio-Sensing	Medical Diagnostics Group	2
Smart Sensing	Structural Monitoring Systems	3

Total number of completed choices: 3

Application Deadline: 15 October 2025

submit clear

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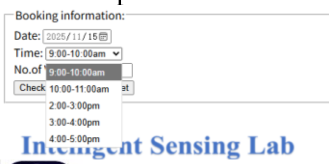
Visit Page

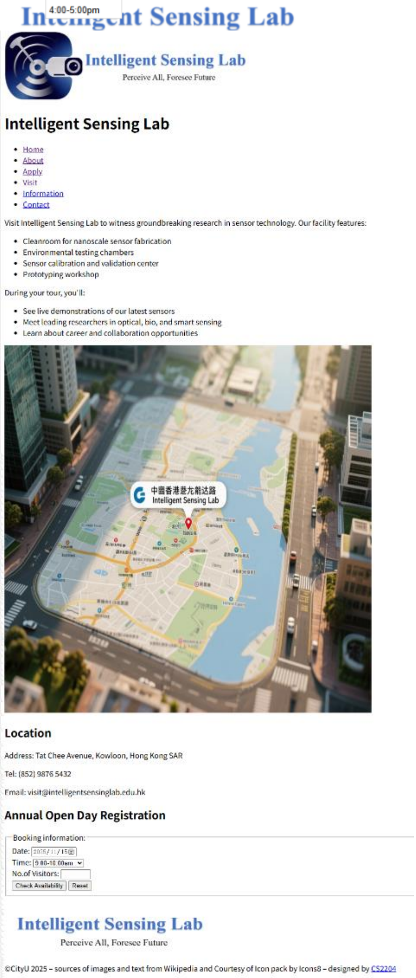
h) This page contains information grouped into six blocks, 3 of which are the standard header, navigation bar, and footer blocks, as introduced above. The remaining three blocks are as follows.

i) **The welcome block** is a block of HTML code containing a paragraph that displays a warm welcome to visitors and introduces the lab and academic environment of the Intelligent Sensing Lab. <p> should be used for each paragraph. You will lose some points if we find all the text in the welcome block crammed into one paragraph.

j) **The location block** contains five types of information as shown in the screenshot: address, telephone, email, and location (**location.png, 800×800**); use the most structured elements to present such information.

k) **The reservation block** is an HTML snippet that contains a reservation form. It displays a form on a web page for booking a visit and collects information about the user's reservation. The user can fill in the information about the reservation to visit the lab, including the date, time, and number of visitors, and submit the form via the submit (Check Availability) button. It would be best to implement everything in the reservation block shown in the screenshot, including but not limited to the legend "Booking information," the form's action attribute, Date, Time, and "No. of Visitors," and Check Availability and reset. You need to note that you should develop at least two different time options. In the screenshot, we have implemented five options.





Information Page

l) This page contains information grouped into five blocks, 3 of which are the standard header, navigation bar, and footer blocks, as introduced above. The remaining two blocks are as follows.

m) There are two **promotion information blocks,** each containing a text passage for the promotion of Intelligent Sensing Lab, followed by a video(**width="700"**) that plays **automatically** and continuously. Inside the <video> tag, use the <source> tag to define the source file for the video. In this project, two source files are provided as alternatives. The browser will select the first supported file type for playback. If the browser does not support either source file type, then the video will not play. The first <source> tag specifies the MP4 file source of the video, and the second <source> tag specifies the WebM file source of the video.

Videos are available on:
https://personal.cs.cityu.edu.hk/~cs2204/2025/video/video1.mp4
https://personal.cs.cityu.edu.hk/~cs2204/2025/video/video1.webm
https://personal.cs.cityu.edu.hk/~cs2204/2025/video/video2.mp4
https://personal.cs.cityu.edu.hk/~cs2204/2025/video/video2.webm

n) The **information table block** contains an HTML table for displaying various research opportunities of different companies and their related information. In the first row of the table, the <th> tag defines the table header, including the column headings "Research Division" "Focus Area" "Open Positions" and "Stipend (HKD/month)". Starting with the second row, each corresponds to information about a division. The text content of each cell is defined using <td> tags. This table presents the various research positions and their related information of different divisions so users can easily compare and find the programs of interest.

Contact Page

o) This page contains information grouped into four blocks, 3 of which are the standard header, navigation bar, and footer blocks, as introduced above. The remaining block is as follows.

p) The contact block creates a table with contact information, including location, address, telephone, and e-mail. Such a table can be used to display the contact information of a school or organization, making it easy for visitors to access relevant information for communication or contact.

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Contact Intelligent Sensing Lab

Department	Address	Tel	Email
Main Laboratory	Tat Chee Avenue, Kowloon, Hong Kong SAR	(852) 9876 5432	info@intelligentsensinglab.edu.hk
Research Partnerships	Tat Chee Avenue, Kowloon, Hong Kong SAR	(852) 9876 5433	partnerships@intelligentsensinglab.edu.hk
Internship Program	Tat Chee Avenue, Kowloon, Hong Kong SAR	(852) 9876 5434	internships@intelligentsensinglab.edu.hk

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Design Page

q) This is an additional free format Web page for you to acknowledge the sources of information used in your design. If you use images, **icons,** or text obtained from the Internet, write the source and links **on** this page. Most information will be put in later for CW2 and CW3. **Whether** you **need something** to acknowledge **or not**, **enter** your name and student ID.

r) This page is mainly used for acknowledgement. Therefore, the TA will only grade this design page based on whether it contains the necessary information, such as your name and acknowledgments (if needed). There is no special formatting requirement for the content of this page, e.g., you can use any suitable tags to present the information in this design page.

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Design Page

Name: CS2204

SID: 12345678

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3. Assessment criteria

You will be assessed by how much and how well you can apply what has been learned from the course; some considerations are:

- **No** styling nor JavaScript is required in CW1 (do not include any style or CSS, except for image sizes added as attributes in the tag as specified in this document; marks will be deducted if your styles confuse our marking)
- use appropriate HTML structural tags (e.g., <header>, <footer>, <div>, <span>), and give as much structure as feasible.
- Adopt all possible good practices discussed in lectures.
- Your website should be organized in proper folders, e.g., HTML and image.
- Make sure the hyperlinks on your web page are available.
- All pages except the design page must pass through validation with no error. You can use https://validator.w3.org/#validate_by_input+with_options for a check.

4. Submission

- Your solution will be submitted to Canvas for grading. Details about the Canvas submission site will be released soon via the Canvas Announcement.

- Submit a zip file of your website with appropriate folders set up so that it can be tested directly by unzip - be careful with your URLs; they should work when your submission is tested as local files on other computers or as Web pages served by a Web server. The TA will download your zip file and unzip it on their own computer and grade it. Therefore, please ensure that the file paths involved in your solution are correct. It is highly recommended to unzip your zip file and test it on another computer to ensure that all pages/images/links are functioning properly before submission.

- DO NOT include video files in your submission file; DO NOT use YouTube video; should use any one given video link in the server folder. Note that this link can only be accessed within the CityU network. If you are outside campus, you should use VPN.

  https://personal.cs.cityu.edu.hk/~cs2204/2025/video/video1.mp4
  https://personal.cs.cityu.edu.hk/~cs2204/2025/video/video2.mp4
  https://personal.cs.cityu.edu.hk/~cs2204/2025/video/video1.webm
  https://personal.cs.cityu.edu.hk/~cs2204/2025/video/video2.webm

  Videos for the promotion information block: video1.mp4, video1.webm, video2.mp4, video2.webm

5. Miscellaneous information

- You should retain a copy of your submission. The same set of web pages will be used again in CW2 and CW3 for adding styles and JavaScript, respectively.
- Some images are available in the Canvas CW1 folder.

~ End ~